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Physiological responses of wheat (*Triticum aestivum* L.) genotypes under drought stress

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Abstract- One of the most significant staple foods for almost two billion people is wheat (*Triticum aestivum* L.). Around the world, wheat is being grown in accordance with different local climates. Drought circumstances include lower or irregular rainfall, uneven rainfall distribution, or increased water demands. Because of the various morphological, physiological, and biochemical alterations at the cellular and molecular levels, wheat's drought response is an extremely complicated process. In this study, the effects of drought stress on two wheat genotypes DBW 187 and HD 2967 at the seedling stage were assessed. Root length, shoot length, RWC, and electrolyte leakage were among the variables where variations were noticeable. In the current study, drought stress results in increased electrolyte leakage across wheat genotypes and decreased root length, shoot length, and RWC. According to the study, wheat genotypes are negatively impacted by drought stress at the seedling stage.

Key words: Wheat, Drought, RWC, Root, Shoot, Proline, MDA, Electrolyte Leakage, MSI

INTRODUCTION

Wheat (*Triticum aetivum* L.) is one of the most important staple food crops of the family Poaceace (Gramineae). It was originated from the southwest Asia and now cultivated worldwide. Wheat provides nearly 55% of the carbohydrates and 20% of the food calories consumed globally.[1] The Indian agriculture system depends on seasonal rainfall occur during the summer monsoon months (June-September) which accounts for 80% of the total annual rainfall of the country. Inter-annual variability of monsoonal precipitation and El Nino-Southern Oscillation (ENSO) affect Indian monsoon and the occurrence of drought significantly. Lower or irregular rainfall, uneven rainfall distribution, or higher water demands are considered drought conditions.[2] Drought poses a higher risk to the food security of India due to its huge population (1.38 billion).[3,4] Wheat exhibits multiple mechanisms to survive under drought stress like drought escape, drought avoidance, and drought tolerance. Shortening the life cycle or growing season of the plant to evade dry ambient conditions is called drought escape, whereas drought avoidance is plant's ability to maintain relatively higher water content in plant tissue in low soil water content. Drought tolerance can be defined as the ability of a plant to maintain its growth and development in scarce water conditions.[5] The drought response of wheat

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is a very complex phenomenon due to the different morphological, physiological, and biochemical changes at cellular and molecular levels. Drought affects leaf size, stem length, and root growth, creates an imbalance in plant-water relations and reduces water use efficiency. Many studies reported physiological changes in cell growth patterns, disturbances in chlorophyll contents, plant-water relations, and photosynthesis occurring in wheat under drought. Morphological changes can be defined as changes in root and shoot systems (effects on height, leaf senescence, flowering, etc.). Biochemical changes take place in different biomolecules and enzymes. Drought stress stimulates a cascade of reactions, in terms of biochemical regulation, i.e. antioxidant production, root deepening, and improved cuticle thickness. The main role of antioxidant production to scavenge reactive oxygen species (ROS) under drought stress conditions has also been recognized in many studies.[6]

In drought conditions, plants have developed osmotic adjustment, one of the fundamental biochemical mechanisms of drought adaptation. Osmotic adjustment changes cell osmotic potential due to the accumulation of osmolytes, thus preserving the physiological functions of the cell in stressful conditions. In addition to their role in osmoregulation, osmolytes have a role in maintaining membrane and other subcellular structures, in reactive oxygen species scavenging, and in gene expression regulation. Depending on the plant metabolism, developmental stage, and environmental conditions, different plants species can accumulate different osmolytes such as amino acids, sugar alcohols, sugars, quaternary amines, and other low-molecular-weight organic solutes.[7] Previous studies showed that proline accumulation contributes to increased osmotic stress tolerance. The expression of the P5CS gene encoding the P5CS enzyme in wheat was shown to be upregulated under osmotic stress and correlated with proline accumulation. Moreover, overexpression of the P5CS gene in transgenic wheat resulted in increased stress tolerance to water-deficit conditions due to increased proline content.[8] Drought causes severe effect on wheat genotypes resulting in physiological and biochemical changes leads to activation of defense mechanism to combat the stress, the effect of drought on wheat genotypes of Eastern India is not much explored hence it becomes important to screen better performing wheat genotype under drought stress.

MATERIALS & METHODS

Plant material, growth conditions and drought treatment: The current study was performed at Dept. of Botany, College of Commerce, Arts & Science, Patliputra University, Patna. The Wheat seeds (HD2967, DBW187) were collected from ICAR-RCER, Patna. Wheat genotypes were grown in hydroponic medium (Hoagland medium) under suitable growth conditions. After six days of germination the wheat seedlings were subjected to drought stress (10% & 20% PEG 6000MW). After stress treatment the samples were harvested. Shoot, Root length of the sample was measured using measuring scale and Relative water content was calculated.[9]

$$\text{RWC (\%)} = \frac{\text{Fresh weight} - \text{Dry weight}}{\text{Turgid weight} - \text{Dry weight}} x\ 100$$

Proline estimation: The plant sample (200 mg) was homogenized in 1 ml of 3% (w/v) sulfosalicylic acid. Homogenate was centrifuged at 10,000g for 15 min. 2 ml of the supernatant, 2 ml of acid ninhydrin and 2 ml of glacial acetic acid were mixed and incubated at 100°C in a water bath for 1 hr. 4 ml toluene was added, and absorbance was taken at a 520 nm.[10]

MDA content was determined through the 2-thiobarbituric acid (TBA) method at 532 nm and then corrected by subtracting non-specific absorbance values at 600 nm by using an extinction coefficient of 155 mM^{-1} cm^{-1}.[11]

Electrolyte leakage and Membrane stability Index: 100 mg plant sample was collected in a 50 ml distilled water. Initial Electrical conductivity was immediately measured. final Electrical conductivity was measured after autoclave at 121°C for 15 min with an EC meter.[12]

Electrolyte leakage = Initial EC/ Final EC × 100

Membrane Stability Index= 1- Initial EC/ Final EC × 100

STATISTICAL ANALYSIS:

Statistical analyses of the data were performed with GraphPad Prism 5.0 (GraphPad Software, Inc.) for complete randomized block design. The experiments were performed in triplicate (n = 3). Data presented as mean ± standard error (SE)

RESULTS & DISCUSSION

Shoot length:

Shoot length was measured in wheat genotypes. Both genotypes showed increase in shoot length subjected to PEG stress. Shoot length of HD2967 showed 22.4, 22.6,

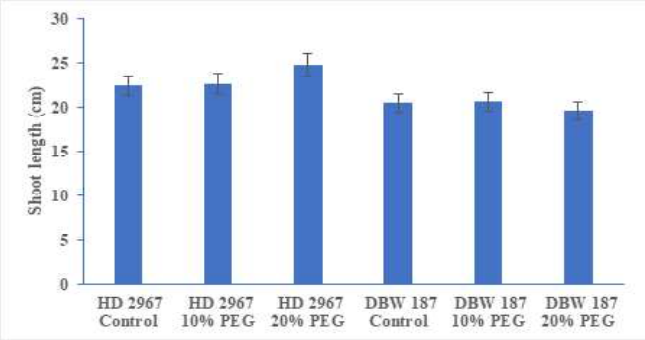


Fig. 1- Shoot length of HD2967 and DBW187 under drought conditions (10% & 20% PEG)

Root length:

Root length was measured in wheat genotypes. Both genotypes showed increase in root length subjected to PEG stress. Root length of HD2967 showed 4.3, 5.3, 6.9 cm while DBW187 showed 5.3, 6, 6.5 cm under Control, 10% and 20% of PEG stress.

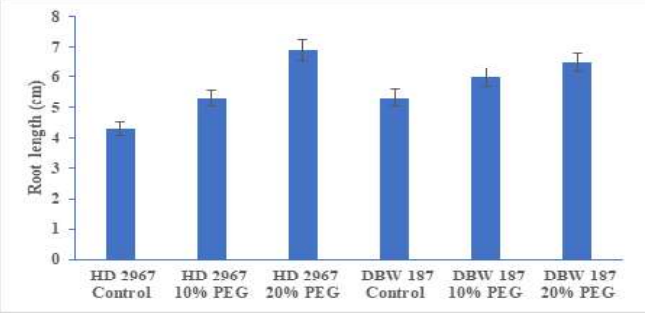


Fig. 2- Root length of HD2967 and DBW187 under drought conditions (10% & 20% PEG)

Relative water content (RWC):

The Relative Water content of whole seedling was measured wheat genotypes. A decrease in RWC was observed in both genotypes. HD2967 showed 93.5, 89.1, 81.7% while DBW187 showed 92.7, 84.5, 70.7% under Control, 10% and 20% of PEG stress.

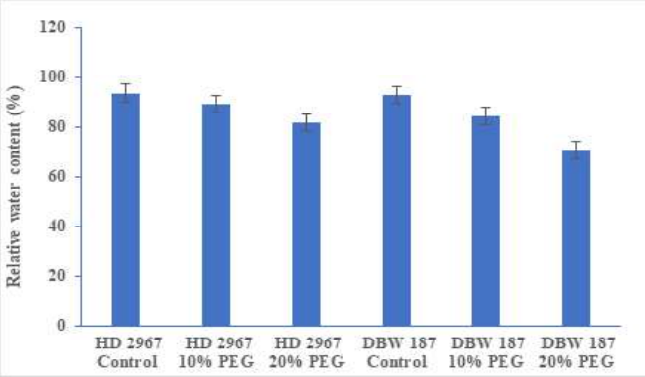


Fig. 3- RWC of HD2967 and DBW187 under drought conditions (10% & 20% PEG)

Proline content:

Proline content was measured in wheat genotypes. Both genotypes showed increase in Proline content subjected to PEG stress. DBW187 showed 1.75, 3.67, 8 µg/mg DW while HD2967 showed 1, 2.9, 6.8 µg/mg DW under Control, 10% and 20% of PEG stress.

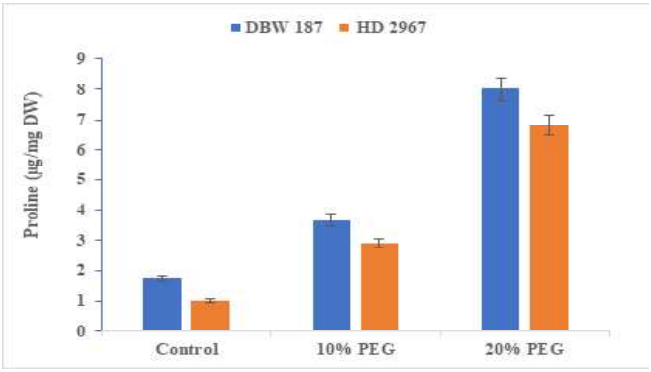


Fig. 4- Proline content of DBW187 and HD2967 under drought conditions (10% & 20% PEG)

Malondialdehyde content (MDA) :

MDA content was measured in wheat genotypes. Both genotypes showed increase in MDA content subjected to PEG stress. DBW187 showed 1.07, 2, 3.89 µg/g FW while HD2967 showed 1.68, 3.79, 5.8 µg/g FW under Control, 10% and 20% of PEG stress.

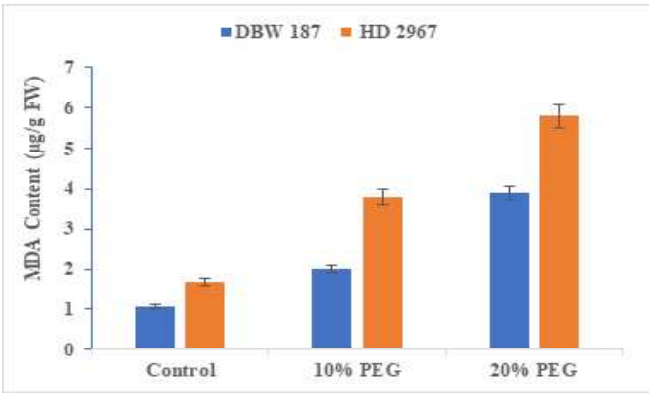


Fig. 5- MDA content of DBW187 and HD2967 under drought conditions (10% & 20% PEG)

Electrolyte leakage (EL):

The Electrolyte leakage of whole seedling was measured in wheat genotypes. An increase in Electrolyte leakage was observed in both genotypes. DBW187 showed 22, 35, 43% Electrolyte leakage while HD2967 showed 24, 40, 49% Electrolyte leakage under Control, 10% and 20% of PEG stress.

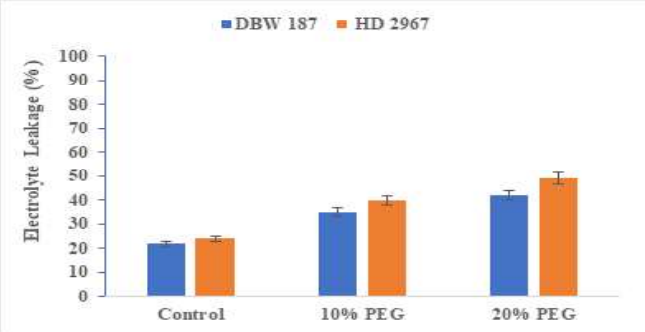


Fig. 6- Electrolyte leakage of DBW187 and HD2967 under drought conditions (10% & 20% PEG)

Membrane stability Index (MSI):

The Membrane stability Index of whole seedling was measured in both wheat genotypes. A decrease in Membrane stability Index was observed in both genotypes. DBW187 showed 78, 65, 58% while HD2967 showed 76, 60, 51% Membrane stability under Control, 10% and 20% of PEG stress.



Fig. 7- Membrane stability Index of DBW187 and HD2967 under drought conditions (10% & 20% PEG)

CONCLUSION

The present study showed DBW187 as better performing wheat genotype in comparison to HD2967 under drought conditions. The findings will be helpful in identification of drought tolerance related genes for understanding their specific role in stress tolerance mechanism.

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